

LALITH KUMAR S

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PROFILE SUMMARY

Electronics and Communication Engineer with a strong interest in Analog Integrated Circuit Design, Semiconductor Devices and Low-Power VLSI Systems. Research experience includes CNTFET-based circuit design, device modeling and performance optimization for nanoscale technologies.

EDUCATION

SRM Institute of Science and Technology <i>B.Tech Electronics and Communication Engineering</i> CGPA: 8.99/10.00	2021 - 2025 Chennai, Tamil Nadu
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Vani Vidyalaya Matric. Hr. Sec. School <i>Higher Secondary Education (Class XII)</i> Percentage: 80.2%	2020 - 2021 Vellore, Tamil Nadu
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Vani Vidyalaya Matric. Hr. Sec. School <i>Secondary Education (Class X)</i> Percentage: 75.0%	2018 - 2019 Vellore, Tamil Nadu
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RESEARCH PUBLICATIONS

Design and Optimization of CNTFET based Binary Edge Activated Synchronous Trigger Flip-Flop (BEAST) 2025

- Published in the 7th International Conference on Inventive Material Science and Applications (ICIMA 2025).
- Designed a CNTFET-based low-power flip-flop architecture for ULSI applications.
- Achieved improvements in power consumption, switching speed and Power Delay Product through device-level optimization.
- Validated performance using Cadence Virtuoso simulations.

CNTFET based Pulse Implied Binary Edge Activated Flip-Flop using Transmission Gate Logic 2025

- Published in the International Conference on Multi-Agent Systems for Collaborative Intelligence (ICM-SCI 2025).
- Proposed a transmission-gate based CNTFET flip-flop architecture for low-power VLSI systems.
- Performed comparative analysis against conventional MOSFET implementations.
- Investigated low-power design methodologies for nanoscale semiconductor technologies.

ANALOG DESIGN EXPERIENCE

Sasken Silicon Technologies Pvt Ltd <i>Analog Circuit Design Engineer Intern</i>	January 2025 - July 2025 Bangalore, Karnataka
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- Designed and analyzed analog building blocks including Amplifiers, Operational Amplifiers, Bandgap References and ADC subsystems.
- Performed DC, AC, transient and corner simulations using Cadence Virtuoso.
- Evaluated trade-offs among gain, bandwidth, power consumption and silicon area through extensive simulation studies.

- Participated in analog circuit validation, design optimization and semiconductor R&D workflows.
- Executed 200+ simulations to characterize circuit performance under various operating conditions.

RESEARCH PROJECTS

CNTFET Device Modeling and Analysis

- Studied the influence of CNT chirality and diameter on threshold voltage, drain current and power consumption.
- Investigated relationships between CNT physical parameters and transistor performance metrics.
- Analyzed device behavior using semiconductor device modeling principles and simulation tools.

Low Power Flip-Flop Design using CNTFET Technology

- Designed and optimized CNTFET-based flip-flop architectures for low-power VLSI applications.
- Evaluated Power Delay Product (PDP), Energy Delay Product (EDP) and power dissipation metrics.
- Performed comparative studies between CNTFET and MOSFET based implementations.

Analog Circuit Design Studies

- Designed and simulated Analog Amplifiers, Operational Amplifiers and Bandgap Reference circuits.
- Analyzed gain, stability, bandwidth and power-performance trade-offs.
- Validated circuit operation through Cadence Virtuoso simulation environments.

TECHNICAL SKILLS

Analog Design: Amplifiers, Operational Amplifiers, Bandgap References, ADC Fundamentals, Memory Elements

Semiconductor Devices: MOSFET Fundamentals, CNTFET Modeling, Device Physics, Semiconductor Fundamentals

EDA Tools: Cadence Virtuoso, Synopsys TCAD, Microwind, MATLAB

Programming: Python, C

Research Areas: Analog IC Design, Low-Power VLSI, Semiconductor Devices, Mixed-Signal Systems, TCAD

ACADEMIC ACHIEVEMENTS

Ranked among the Top 5 students in the Department of Electronics and Communication Engineering.

Published two international conference papers in CNTFET-based VLSI design.

Conducted research in low-power CNTFET-based digital circuits and semiconductor device modeling.